

**Polymeric Osteoconductive Bone scaffolds : Synthetic and natural polymers
used for bone regeneration**

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Abstract

Although bones are highly crucial organs that provide structural and functional support for the human body, their vulnerability to various diseases and defects and the resulting loss of bone has raised the necessity to formulate solutions to help regenerate them. One of the main objectives of bone tissue engineering is to address the limitations of the current treatment methods through bone scaffolds based on various materials. Among osteogenic (cellular bone growth), osteoinductive (bone cell generating) and osteoconductive (bone growth on a surface) functions of the bone tissue engineered-treatments, this review focuses on osteoconductive treatments as they involve many biomaterial-based approaches. The paper provides a brief history on osteoconductive scaffolds, subsequently introducing key physicochemical properties of polymeric bone scaffolds. This is followed by brief summaries on key synthetic and natural polymers which addresses their advantages and limitations based on the physicochemical properties. The review specifically sheds light on the commonly used synthetic and natural polymeric biomaterials used for bone regeneration in the 2010s and onward. This review aims to provide a general overview of the development of the bone tissue engineering field in osteoconductive polymer biomaterials while introducing recent efforts to improve and optimize their properties.

Key Words: Osteoconductive, Biomaterials, Bone tissue engineering, natural and synthetic polymer, bone regeneration

Introduction

Bone, a highly specialized connective tissue, plays a crucial role in not only providing structural support, protecting vital organs, and enabling mobility but also as a calcium storage and blood cell production site (1). However, its vulnerability to damage or loss due to disease, trauma and hormonal imbalances presents significant challenges for bone repair and regeneration (2,3). With over 400 million of the world population experiencing symptoms from bone defects, the need for effective clinical solutions like bone scaffolding for bone regeneration and repair is vast (4).

Bone tissue engineering is an emerging field that seeks to address the limitations of traditional treatments for bone defects. The gold standard bone scaffolds are autogenous, patient-derived bone grafts (autografts) associated with low rejection or disease transmission rate, yet they have limited sources, and often pose surgical risks. Allogeneic bone grafts (allografts) from other donors and xenogeneic bone grafts (xenografts) from other species have

the risks of immune rejection and requiring sterilization (2). Bone allografts can use natural materials like bone tissue, powder or collagen or synthetic materials like calcium phosphate or calcium sulfate to repair badly damaged bones. Due to the issues with each type of bone graft, bone scaffolds began to be introduced. A bone scaffold is a structure which has properties which help repair bone. Certain bone scaffolds also allow cells to grow into the scaffold in order to integrate and rebuild the bone. These scaffolds are known as osteoconduction. To address the limitations associated with the existing bone grafts, biomimetic bone scaffolds, serve as temporary frameworks with tunable porosity and degradation rates to guide bone formation (5). Recent advancements in scaffold design have integrated novel materials and biologically active components that drive osteogenic, osteoconductive or osteoinductive properties (2).

Osteoconduction refers to the three-dimensional process of bone growth on a porous surface or structure (6). Applying this principle, osteoconductive scaffolds can be designed with porous microstructures providing the spatial advantage for the recruitment and expansion of human mesenchymal stem cells (hMSC), osteoblasts, osteoclasts and angiogenic factors required for the osteoinductive, in other words, bone-building processes (7). Recent therapies have shown advancements in incorporating synthetic and natural polymers such as poly(lactic-acid) (PLA), poly(glycolic-acid) (PGA), and collagen, to bone scaffolds with improved functionalities. Among these materials, polymers are one of the most widely used material classes for bone scaffolds due to their versatility and ability to be tailored for specific needs in bone repair. They offer high biocompatibility and a broad range of tunable properties, including environmental sensitivities, mechanical characteristics, and adjustable degradation rates, which make them ideal for a variety of clinical applications (9)(10). This review focuses on different types of currently used natural and synthetic polymeric material for osteoconductive bone scaffolds with brief historical context regarding the development of the field in the 1970s. Specifically, the review will introduce various polymeric bone scaffolding materials and their advantages and limitations citing peer-reviewed articles. Through this review, we aim to explore the applications of important physicochemical properties and developments of the osteoconductive polymeric bone scaffolds used for bone regeneration.

Brief historical background of the development of osteoconductive scaffolds

Bone transplants, a procedure practiced for over a century, utilize autograft, allograft, or xenograft bone scaffolds. As early as the late 19th century, researchers observed that transplanted bone degraded over time and was replaced by the growth of new bone. This discovery led to the realization that preserving the structure of the damaged bone could enable

natural healing without requiring a full transplant (Axhausen, 1909). Building on this insight, osteoconductive bone scaffolds—porous structures implanted at or near bone defects or alongside transplanted bone needing additional support—emerged as a widely adopted treatment option (Ghayor & Weber, 2018).

There was a wide gap between this realization in the late 19th century and the use of osteoconductive scaffolds in vivo (Mizuno & Hirohata, 1984). It wasn't until the 1970s that some of the first clinical usages of bone scaffolds were reported. In 1978, Parrish et al. published a paper discussing a particular osteoconductive scaffold made of Marlex Mesh which fibroblastic tissue could migrate into (Parrish et al. 1978). This was quickly followed by development of scaffolds composed of hydroxyapatite, the main inorganic building block of the bone. These hydroxyapatite scaffolds had a range of pores that allowed for osteoconduction (Sartoris et al. 1986) (Davis et al, 1990). After this point bone scaffolds only progressed further to meet the goal of faster and more extensive bone repair with the least possibility of triggering the immune response in the patient's body for both the recipient or the donor site of the bone graft.

Important physicochemical properties of polymeric bone scaffolds

Polymer scaffolds serve as a matrix that recreates the bone tissue microenvironment, supporting bone cell growth and function. Therefore, their physicochemical properties become important to ensure proper cell adhesion, recruitment, proliferation and differentiation. Important properties of successful bone scaffolds include biological properties like biocompatibility, bioactivity and biodegradability and mechanical properties like compressive strength and Young's modulus . Polymer biocompatibility is influenced by factors such as molecular chemistry, molecular weight, solubility, shape, and structure of the implant (9). Additionally, key material characteristics like hydrophilicity and hydrophobicity play crucial roles in determining a polymer's compatibility with the body (9). The biocompatibility of the polymer scaffold also heavily relies on its biodegradability, influenced by the hydrolytic or enzymatic sensitivity of the molecular bonds between the polymer molecules (10). Bioactivity of the polymeric material, such as its response to various physiologically present ions and compounds, could affect the osteoconductive properties.

Next, having similar mechanical properties such as the compressive strength and Young's modulus as the bone is crucial for the function of the bone scaffolds. Native compact bone, also called cortical bone, shows high Young's modulus of 15-20 GPa with compressive strength of around 100-230 MPa, while having low porosity of 5-30%. On the other hand, the

native trabecular bone, also called spongy bone due to the high porosity of 30~95%, its Young's modulus is 0.1-2 GPa while the compressive strength is 2-12 MPa (10). Therefore, it is important to consider these properties of the native bone when designing the bone scaffold to be inserted into the bone. Additional mechanical properties include porosity, elasticity and fatigue strength, that resemble that of the natural bone. Especially, having optimal porosity is important as cell diffusion and migration happens through interconnected pores of the material. For instance, having pore size greater than 300 μm and over 90% porosity is highly recommended to ensure mechanical stability, vascularization, nutrient diffusion and cell-biomaterial interaction with adequate surface area (17). However, studies have shown that smaller pores (nano or micro-sized pores) enhance the mechanical strength of the scaffold matrix, therefore leading to high structural durability while still maintaining the similar level of biological performance (18). Therefore, the balance between porosity and mechanical strength must be carefully considered depending on the applications.

Types of polymers - Synthetic and Natural

The properties of polymers are heavily influenced by whether they are synthetic or natural, as each has their own unique characteristics. Natural polymers tend to be bioactive and biomimetic but pose risks such as the triggering of immune responses, susceptibility to microbial contamination, limited mechanical strength, and less controlled degradation (10).

On the other hand, synthetic polymers can be engineered to have specific chemical properties that enable desirable mechanical characteristics and controlled degradation rates (9). These polymers also have batch to batch uniformity. However, synthetic polymers lack biomimetic properties.

It is important to note that no single polymer possesses all the ideal attributes needed for a bone scaffold. Each polymer has its own advantages and disadvantages, which must be carefully considered when selecting the best material for a specific patient's needs and the type of bone repair required. This is also the reason why many bone scaffold applications use copolymer systems where they adapt advantages of various polymers to improve the systematic performance as a bone scaffold and promote bone regenerations ().

Synthetic polymers

To ensure that the synthetic polymer aligns optimally with the patient's needs, several options are available, each offering distinct properties. The main types of synthetic polymers used in medical and biomedical applications will be briefly described below.

Polyglycolide (PGA), Polylactide (PLA) and Poly(lactide-co-glycolide) (PLGA)

Polyglycolide (PGA) and Polylactide (PLA) are first-generation biodegradable synthetic polymers widely used in early biomedical applications, particularly in orthopaedic medicine and bone tissue engineering (Pillai & Sharma, 2010; Athanasiou et al., 1998). Although largely replaced by poly(lactide-co-glycolide) (PLGA) in research after the late 2000s, understanding the unique characteristics of PGA and PLA is critical to appreciating their early role as bone scaffold materials (Silverberg et al., 1986).

The mechanical properties and processability of PGA and PLA, attributed to their crystallinity, make them suitable for applications such as bone scaffolds. In terms of the crystallinity, PLA typically has a crystallinity of 35-40%, contributing to moderate mechanical strength compared to PGA. In contrast, PGA demonstrates higher crystallinity (44-55%) and stereoregularity, resulting in superior tensile strength and mechanical stability (Chujo et al., 1967). Additionally, PGA is insoluble in organic solvents, making it advantageous for processing and storage (Pillai & Sharma, 2010).

PLGA, a copolymer of the polyesters PLA (Poly(lactic Acid)) and PGA (Poly(glycolic Acid)), has largely replaced the usage of PLA and PGA in bone scaffold materials. The structural integrities and degradation weights of PLGA can be tuned and improved by combining PLA and PGA in different ratios (). Also, both PGA and PLA offer good biocompatibility and biodegradation as they consist of multiple repeating units of glycolic acid and lactic acid monomers. These are natural metabolites, making PLGA bioinert and useful for temporary biomedical applications (Toosi et al., 2019). They degrade spontaneously through hydrolysis of ester linkages connecting the monomer chains consisting of the polymer backbone (Middleton & Tipton, 2000).

However, both PGA and PLA have limitations of lacking osteoconductive surface properties that are crucial for enhancing cell adhesion and proliferation, as well as the low solubility restricting broader applications (Pillai & Sharma, 2010). Also, the lack of mechanical strength has made using PLGA alone as a bone scaffold difficult. . Additional limitations of PLGA include the possible accumulation of acidic byproducts from degradation that could lead to immunogenic responses or structural failure (). Integration of PLGA with another polymer like polyphosphazene can be used to compensate for the effect of the acidic environment (9).

To address these issues, PLGA is used in combination with other materials for scaffolding. In 2020, Xie et al. created Attapulgite-doped electrospun fibrous PLGA scaffolds (). In the study, the PLGA composite could leverage its favorable cell adhesion without the limitation of PLGA's limited mechanical strength. Also, Go et al. and Lee et al. have combined PLGA polymer with beta-tricalcium phosphates (β -TCP) and magnesium oxide (MgOH_2) to generate osteoconductive, anti-inflammatory scaffolds for bone repair and regeneration (Go et al. 2020; Lee et al. 2019). The strategy of adding the inorganic compounds improved the limited mechanical properties and alleviated inflammatory response. These strategies enable PLGA to have superior biodegradability and biocompatibility, making it a promising osteoconductive bone scaffolding material.

Poly(L-lactic acid) (PLLA) and Poly(DL-lactide) (PDLLA)

Poly(L-lactic acid) or PLLA is a crystalline polymer with variable crystallinity depending on factors such as molecular weight and processing conditions (Wang & Mano, 2005). A higher molecular weight PLLA scaffold is likely to have lower crystallinity which causes decreased strength (Makadia & siegel, 2012). Cooling and heating processings are also incredibly important for determining the percent crystallinity of the polymer and can be leveraged to increase or decrease the mechanical strength of PLLA scaffolds ().

At its average crystallinity, PLLA has a high modulus of elasticity, roughly 4.8 GPa, and good tensile strength, around 65 MPa, both of which allow it to be used for load bearing applications like bone scaffolding (9). However this strength degrades much faster than the material itself. While the strength begins to decrease roughly 6 months after implantation, the actual scaffold itself can take more than 5 and a half years to fully reabsorb (). This full reabsorption time can be modified by changing the percent crystallinity or its porosity (). On another note, PLLA degradation, along with biocompatibility, can both be improved by copolymerization ().

These modified PLLA scaffolds not only have increased strength, biocompatibility and degradation, but it is a relatively hydrophobic material which increases its potential to be used for an osteoconductive bone scaffolding material (). In 2018, Ge et al. combined PLLA with strontium and Hydroxyapatite in order to create an even more hydrophobic porous bone scaffold that showed enhanced osteoinduction and increased protein absorption and compressive strength ().

Poly(DL-lactide) (PDLLA) is a variation of PLLA, but with a more amorphous morphology (9). Since it is the crystalline structure of PLLA that gives the material its strength, PDLLA's amorphous structure means that it has a relatively lower strength at 1.9 GPa (9). A benefit of the amorphous structure of the PDLLA is that it causes a much faster degradation rate (). This can be useful for drug eluting scaffolds and is often leveraged when combining PDLLA with other materials to increase the strength of the scaffold while keeping high degradation rates. For example, Han et al. created a bioactive glass and PDLLA scaffold that they loaded with Dimethyloxallylglycine (DMOG) in order to promote bone repair (). In this case, the scaffold showed promise with the release of DMOG, promoting cell proliferation, adhesion and differentiation when combined with other materials to make an osteoconductive and osteoinductive scaffold, increasing patient bone repair ().

Polycaprolactone (PCL)

Polycaprolactone (PCL) is an aliphatic, semi-crystalline polyester that was first investigated because of its low cost and high processability (). PCL is commonly seen in 3D printed scaffolds due to its high processability (). Other advantages of PCL include its high solubility in organic solvents and its ability to form composites with different materials, reducing some of its limitations ().

One of these limitations is PCL's relatively slow degradation rate. While slow degradation rates do allow time for bone to heal, too long a time can start to cause complications. Typically, it takes between 2 to 3 years for an unmodified PCL scaffold to degrade, while bone can mend in a couple of months (). Like deal with this, PCL can be modified to increase its degradation rate by increasing the amorphous regions of the material, which can be done by creating co-polymers with DL-lactide ()(). Another limitation of PCL is its mechanical strength, which is about 23 MPa, which is below the normal strength needed for a working bone scaffold (100-150 MPa) (9). To overcome this limitation PCL composite scaffolds are produced (). One example of a PCL composite scaffold can be seen in Jan et al., where PCL is combined with collagen to produce an osteoconductive bone scaffold with mesenchymal stem cells (MSCs) (). PCL was also used by Remy et al. where gelatin coated, 3D printed PCL scaffolds were used to increase bone repair (). Remy et al. illustrated the major benefit of PCL, its processability, which allows it to be 3D printed into numerous shapes, allowing for individualization of the bone scaffolds.

Polyanhydrides and Crosslinked Polyanhydrides

Polyanhydrides are degradable polymers with limited mechanical strength in their unmodified form. To address the mechanical strength limitations of polyanhydrides, poly(anhydride-co-imides) were developed. These polymers used imide segments in the backbone to enhance their strength (9). In studies by Laurencin et al., the mechanical performance and biocompatibility of poly(anhydride-co-imides) were evaluated for their abilities to act as bone tissue regeneration scaffolds (9). This research demonstrated that poly(anhydride-co-imides) could exhibit compressive strengths of 50-60 MPa, which would be tough enough to act as a bone scaffolding material (). Additional studies like those of Dasgupta et al. leveraged the polyanhydride properties to create an injectable paste that acted like a bone replacement ().

An alternative approach to enhance the mechanical properties of polyanhydrides is to make it injectable and photo-crosslinkable (). This improves their applicability to irregularly shaped bone defects (9). On top of the new properties, these crosslinked polyanhydrides also have the lack of immunogenic response that is associated with polyanhydrides. However, the cross-linked version still has lower range compressive strength, 30-40 MP, according to Langer et al. (). The strength would need to be further modified in order to deal with bones with high modulus, but for injectable bone scaffolds polyanhydrides are effective and useful.

Co-polyester poly(propylene) fumarate (PPF)

Co-polyester poly(propylene) fumarate (PPF) is a cross-linkable, biodegradable, and high-strength polymeric biomaterial (). PPF is a degradable synthetic polymer with several factors—such as molecular weight, types of curing agents, and crosslink density—affecting the degradation time. The products of these degraded scaffolds are all benign which leads to PPF scaffolds having a higher level of biocompatibility. ().

The main use for PPF is taking advantage of its ability to form injectable solutions, which can be set in vivo and be molded into the desired shape. This adds to its versatility since it can be used for bone impairments with uncommon shapes. In Fang et al. They leveraged the injectability of PPF to treat simulated metastatic lytic defects with greater success than the commonly used PMMA and with significant improvement over no formal treatment (). Despite the lower mechanical strength, the injectability, biocompatibility, and biodegradability of PPF allowed it to be considered a promising candidate for osteoconductive bone scaffolding.

Polyphosphazene

Polyphosphazene polymers are a class of inorganic-organic hybrid materials that have existed since the mid-1960s, with biodegradable variants having only been developed more recently (). Specialized polyphosphazenes, particularly those incorporating specific side groups, have been extensively explored for bone regeneration applications (). These side groups were deliberately introduced to create specific characteristics, such as modulating and controlling degradation rates, hydrophobicity or hydrophilicity, and other mechanical properties (). For example the degradation rate is correlated with the polyphosphazene content in the scaffold allowing for controlled degradation rates for Polyphosphazene bone scaffolds().

Polyphosphazenes can have multiple different designs, due to the incorporation of various side groups, making them particularly attractive for bone scaffolding (). For example, incorporating the ethyl glycinate group enhances cell adhesion and growth, and accelerates the degradation rate (). In 2014, Rothemund et al. used polyphosphazenes as a scaffolding material and demonstrated their biocompatibility, and cell adhesion and proliferation abilities (). This biocompatibility and biomimetic activity along with the control of degradation rates gives Polyphosphazenes great potential for use as a bone scaffolding material.

Natural polymers

While synthetic polymers offer a wide range of customizable characteristics for bone scaffolds, naturally occurring polymers provide enhanced biocompatibility and bioactivity, making them a compelling alternative (). Natural polymers include polysaccharides and proteins and some of these are similar to or direct components of the extracellular matrix (ECM)(). By selecting polymers that resemble the ECM, toxicity and immune responses are minimized, increasing the likelihood of creating an effective scaffold.

Collagen

Collagen is the most abundant protein in mammals and serves as the primary structural component of musculoskeletal tissues (). The primary structure of collagen consists of repeating triplets of (Glycine-X-Y)_n (). This repeating sequence imparts a helical structure to the protein, contributing to its mechanical strength and flexibility (). Other benefits of collagen are its degradability and processability, ability to be shaped into various forms ().

One of the unique benefits of collagen is its ability to initiate and propagate mineralization—the process of bone formation—independently, unlike many other ECM-like materials (). Collagen also allows for crosslinking using various agents which offer different improved mechanical properties (9).

Collagen-based biomaterials are widely used in bone scaffolds due to their ability to support tissue regeneration (). For instance, a collagen-based composite of fibrillar collagen, hydroxyapatite, and tricalcium phosphate (Collagraft) is FDA-approved for use as a biodegradable synthetic bone graft substitute and demonstrates favorable interactions with proteins, promoting bone healing ().

Despite its beneficial properties, collagen-based biomaterials have some drawbacks. These include mild immunogenicity due to antigenic sites (). Other challenges include its high cost, variations in its physicochemical and degradation properties, a heightened risk of infection, stringent processing conditions, and the hydrophilicity of collagen, which can cause swelling in vivo ().

Hyaluronic acid (HA)

Hyaluronic acid is a glycosaminoglycan but causes limited protein aggregation and is susceptible to degradation (). HA plays a role in improving the mechanical strength of the polymeric matrix. For example, HA-based gelatin hydrogel scaffolds, combined with biphasic calcium phosphate, had improved interconnectivity in the pore network and displayed a high compressive strength of 2.8 MPa ().

Its susceptibility to degradation, lack of protein aggregation, and improved mechanical strength makes it a promising candidate for bone scaffolding ().

While HA passively serves as a structural component of connective tissues, it is also actively involved in modulating cell migration and differentiation during embryogenesis (). Additionally, HA plays a crucial role in regulating ECM organization, metabolism, and processes like metastasis, wound healing, and inflammation (). HA's benefits are further enhanced when crosslinked. Crosslinking reduces its degradation rate, stabilizing the structure and extending its functionality (). Crosslinked HA supports angiogenesis and without unwanted immune responses. Recent studies have also confirmed that N-cadherin-modified hyaluronic acid promotes bone formation and enhances human mesenchymal stem cell osteodifferentiation (). Its contribution to bone healing functions both when crosslinked and on its own are key contributing factors to its use in bone scaffolding.

Despite its numerous benefits, HA does have some drawbacks. Harsh chemicals and high temperatures used in certain processing methods can limit its bioactivity, reducing its processability (). Additionally, HA's strength can vary depending on the processing method,

leading to potential material inconsistencies (). Nonetheless, the numerous advantages of hyaluronic acid make it a material worth further exploration. Efforts to overcome its limitations will be crucial, as HA holds great promise for use in bone scaffolding.

Discussion.

The field of bone tissue engineering is continuously developing. As technology advances, we will only continue to see improvements in not only the success of certain polymers in bone tissue regeneration but also the expansion of different methods and strategies that can be used beyond the field of bone tissue engineering. This review focused on the advantages and disadvantages when using different bone scaffolding polymers.

Polymers were the focus of the review, as they have many favorable traits due to their dynamic nature, offering advantageous mechanical characteristics, biocompatibility, and ease in adjusting the porosity and degradation of polymer scaffolds. The polymers discussed were strategically differentiated into two categories of synthetic or natural polymers. The use of one polymer over the other is situational and patient-dependent; There were many ways discussed that went over the numerous “not one-size fits all”, as each polymer has its unique advantages and disadvantages. Some of the methods mentioned included the mixing of polymers, to get a “best of both worlds effect”, using one polymer for its favored advantageous effects and making up for its disadvantages with another polymer that addresses those aspects, creating different co-polymer combinations that also allowed for the customization of factors such as stability, porosity, and degradability. A different kind of popular combination is the incorporation of bioceramics such as bioglass, hydroxyapatite, and tricalcium phosphate, and the incorporation of stem cells, like MSCs, is on the rise and an area that has a promising potential but still has lengths to go before we can see the normalization of it in the clinical field.

An important aspect, not as focused in the field, that has an equal or greater importance than the development of new novel methods, is the integration of developmental methods into clinical practice. To put this into perspective, “less than 5% of life science scientific discoveries originated in academia will successfully transition into a change in clinical practice, new medications, diagnostics, or devices” (Montoya et al. 2021) showing us how little we are benefiting off scientific advancements. In the field of bone polymer scaffolding and even more generally, in the biomedical technology field, not only do we need more research to get closer to our goal of the perfect bone scaffold, but we also need a system put in place that can more readily take revolutionary technologies from bench to bedside, giving them the push they need to get into clinical testing to access their true potential, allowing for a greater number of

advancements in healthcare and allowing us to overcome the so-called “valley-of-death” (Montoya et al. 2021). Governmental leadership is essential to establish academic, regulatory, and industry frameworks that facilitate the integration of these technologies into society—a process that has been delayed for far too long (Montoya et al. 2021).

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